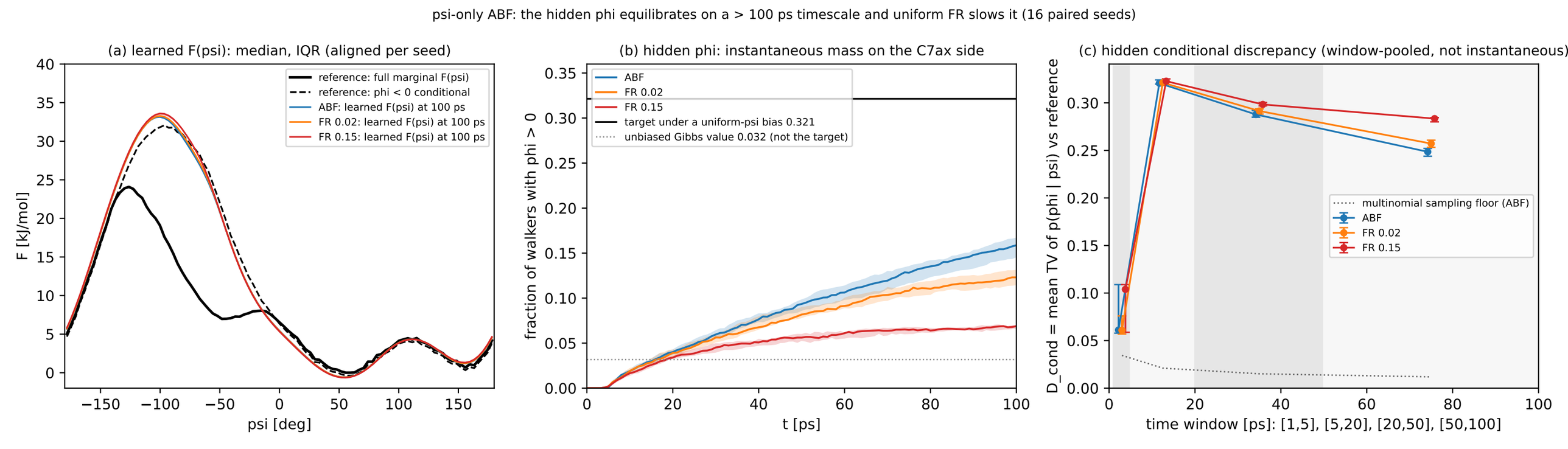
psi-only ABF: the hidden phi equilibrates on a > 100 ps timescale and uniform FR slows it (16 paired seeds)
(a) learned F(psi): median, IQR (aligned per seed)
reference: full marginal F(psi)
reference: phi < 0 conditional
ABF: learned F(psi) at 100 ps
FR 0.02: learned F(psi) at 100 ps
FR 0.15: learned F(psi) at 100 ps
F [kJ/mol]
psi [deg]
(b) hidden phi: instantaneous mass on the C7ax side
ABF
FR 0.02
FR 0.15
target under a uniform-psi bias 0.321
unbiased Gibbs value 0.032 (not the target)
fraction of walkers with phi > 0
t [ps]
(c) hidden conditional discrepancy (window-pooled, not instantaneous)
multinomial sampling floor (ABF)
ABF
FR 0.02
FR 0.15
D_cond = mean TV of p(phi | psi) vs reference
time window [ps]: [1,5], [5,20], [20,50], [50,100]